

Seismology Problem Set 1

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1. Within Lecture 13, we derived the following equation

$$\rho \frac{\partial v_i}{\partial t} - \frac{\partial T_{ij}}{\partial x_j} = 0,$$

for an elastic body. Here ρ is the referential density, $v_i = \frac{\partial \varphi_i}{\partial t}$ the velocity with φ the motion, and $\mathbf{T}$ the first Piola–Kirchhoff stress tensor.

- (a) For an arbitrary sub-body, U , of the reference body, M , show that the equality

$$\frac{d}{dt} \int_U \rho v_i d^3\mathbf{x} = \int_{\partial U} T_{ij} n_j dS,$$

holds, where ∂U is the boundary of U . What is the physical significance of this result? In particular, what does the vector $T_{ij} n_j$ represent?

- (b) The angular momentum of the sub-body U about the origin is given by

$$\int_U \rho \epsilon_{ijk} \varphi_j v_k d^3\mathbf{x}.$$

Show that the angular momentum of every such sub-body changes only through the torque exerted by the tractions on its boundary if and only if $\mathbf{FT}^T$ is a symmetric tensor. Defining the second Piola–Kirchhoff stress, $\mathbf{S}$, through $\mathbf{T} = \mathbf{FS}$, conclude that $\mathbf{S}$ is symmetric.

2. Continuing from the previous question, we define the energy density of the body by

$$E = \frac{1}{2} \rho v_i v_i + W(\mathbf{x}, \mathbf{F}),$$

where W is the strain energy function, so that $T_{ij} = \frac{\partial W}{\partial F_{ij}}$. Show that

$$\frac{\partial E}{\partial t} + \frac{\partial s_i}{\partial x_i} = 0,$$

where s_i is a vector field to be determined. By integrating this identity over an arbitrary sub-body, U , explain the physical meaning of s_i , and hence show that the total energy of the body is conserved.

3. The Lagrangian density for the linearised motion of an elastic body subject to a stress glut is given by

$$\mathcal{L} = \frac{1}{2} \rho \frac{\partial u_i}{\partial t} \frac{\partial u_i}{\partial t} - \frac{1}{2} A_{ijkl} \frac{\partial u_i}{\partial x_j} \frac{\partial u_k}{\partial x_l} + \bar{S}_{ij} \frac{\partial u_i}{\partial x_j},$$

where $\mathbf{u}$ is the displacement vector, A_{ijkl} the elastic tensor, and $\bar{\mathbf{S}}$ the stress glut. Using this expression, write down the equations of motion and natural boundary conditions; here you may assume the standard form of the Euler–Lagrange equations.

4. The principle of material frame indifference shows that the strain energy function, W , of an elastic body depends on the deformation gradient, $\mathbf{F}$, only through the right Cauchy–Green deformation

tensor, $\mathbf{C} = \mathbf{F}^T \mathbf{F}$, and hence we can write $W(\mathbf{x}, \mathbf{F}) = U(\mathbf{x}, \mathbf{C})$, for some auxiliary function, U . The elastic tensor, A_{ijkl} , is defined by

$$A_{ijkl} = \frac{\partial^2 W}{\partial F_{ij} \partial F_{kl}},$$

where the partial derivative on the right-hand side is evaluated at the equilibrium value $\mathbf{F} = \mathbf{I}$. Assuming that the equilibrium is stress-free, show that the elastic tensor possesses the symmetries $A_{ijkl} = A_{jikl} = A_{ijlk} = A_{klij}$. How many independent components does such an elastic tensor have? Do these symmetries remain if there is a non-zero equilibrium stress?

5. In lectures we found the stress glut for an idealised fault surface Σ to be $\bar{S}_{ij} = A_{ijkl} w_k \hat{n}_l \delta_\Sigma$, where $\mathbf{w}$ is the displacement discontinuity vector, which is tangential to Σ , $\hat{\mathbf{n}}$ is the unit normal to Σ , and δ_Σ is the Dirac surface distribution for Σ .

- (a) Consider a planar fault with unit normal $\hat{\mathbf{n}}$ within an isotropic body. Show that

$$\bar{S}_{ij} = \mu(w_i \hat{n}_j + w_j \hat{n}_i) \delta_\Sigma,$$

and comment on the absence of λ . Verify that this stress glut is symmetric and has zero trace.

- (b) Consider the linearised equations of motion in a body M subject to a body force f_i and an applied traction t_i on ∂M ,

$$\rho \frac{\partial^2 u_i}{\partial t^2} - \frac{\partial}{\partial x_j} \left(A_{ijkl} \frac{\partial u_k}{\partial x_l} \right) = f_i \quad \text{in } M, \quad A_{ijkl} \frac{\partial u_k}{\partial x_l} \hat{n}_j = t_i \quad \text{on } \partial M,$$

with the body at rest prior to $t = 0$. The solution of this problem can be expressed in terms of a Green function, $G_{ij}(\mathbf{x}, t; \mathbf{x}', t')$, in the form

$$u_i(\mathbf{x}, t) = \int_0^t \int_M G_{ij}(\mathbf{x}, t; \mathbf{x}', t') f_j(\mathbf{x}', t') d^3 \mathbf{x}' dt' + \int_0^t \int_{\partial M} G_{ij}(\mathbf{x}, t; \mathbf{x}', t') t_j(\mathbf{x}', t') dS' dt',$$

where the detailed form of G_{ij} will not be needed. Recall that a stress glut, $\bar{S}_{ij}$, enters the linearised equations of motion as an effective body force $f_i = -\partial \bar{S}_{ij} / \partial x_j$ together with an effective traction $t_i = \bar{S}_{ij} \hat{n}_j$, where $\hat{\mathbf{n}}$ is the outward unit normal to ∂M . Note that within the integrals above f_j , t_j and $\hat{\mathbf{n}}$ are all evaluated at the integration point $\mathbf{x}'$. Show that the resulting displacement can be written as a single integral over the stress glut,

$$u_i(\mathbf{x}, t) = \int_0^t \int_M \frac{\partial G_{ij}}{\partial x'_k}(\mathbf{x}, t; \mathbf{x}', t') \bar{S}_{jk}(\mathbf{x}', t') d^3 \mathbf{x}' dt'.$$

- (c) Using the result of part (a), show that the displacement generated by slip on a planar fault within an isotropic body is

$$u_i(\mathbf{x}, t) = \int_0^t \int_\Sigma \mu \left[\frac{\partial G_{ij}}{\partial x'_k} + \frac{\partial G_{ik}}{\partial x'_j} \right] (\mathbf{x}, t; \mathbf{x}', t') w_j(\mathbf{x}', t') \hat{n}_k(\mathbf{x}') dS' dt',$$

and comment on the physical content of this expression. (The normal $\hat{\mathbf{n}}$ is constant for a planar fault, but it has been written as a function of $\mathbf{x}'$ because the result holds unchanged for a general fault surface.)

6. Consider plane wave propagation in a homogeneous and transversely isotropic whole space. For such a material the elastic tensor takes the form

$$A_{ijkl} = (A - 2N) \delta_{ij} \delta_{kl} + N(\delta_{ik} \delta_{jl} + \delta_{il} \delta_{jk}) + (F - A + 2N)(\delta_{ij} \hat{v}_k \hat{v}_l + \hat{v}_i \hat{v}_j \delta_{kl}) \\ + (L - N)(\hat{v}_i \hat{v}_k \delta_{jl} + \hat{v}_j \hat{v}_k \delta_{il} + \hat{v}_j \hat{v}_l \delta_{ik} + \hat{v}_i \hat{v}_l \delta_{jk}) + (A + C - 2F - 4L) \hat{v}_i \hat{v}_j \hat{v}_k \hat{v}_l,$$

where A , C , F , L and N are elastic moduli, and $\hat{\mathbf{v}}$ is a unit vector along the symmetry axis. Let θ denote the angle between the propagation direction, $\hat{\mathbf{p}}$, and the symmetry axis, and regard the departure of the medium from isotropy as small.

- (a) Using first-order perturbation theory, obtain an expression for the phase speed of a quasi P-wave as a function of θ .
- (b) Using degenerate perturbation theory, obtain the first-order phase speeds of the two quasi S-waves as functions of θ , and describe their polarisations. Check your results against the exact phase speeds for propagation along and perpendicular to the symmetry axis. [Hint: choose one of the unperturbed polarisation vectors to lie in the plane containing $\hat{\mathbf{p}}$ and $\hat{\nu}$, and the other perpendicular to this plane.]
7. Show that the time-dependent Schrödinger equation for a free particle

$$i\hbar \frac{\partial \psi}{\partial t} = -\frac{\hbar^2}{2m} \nabla^2 \psi,$$

admits plane wave solutions of the form

$$\psi(\mathbf{x}, t) = a \exp \left[\frac{-i(Et - \mathbf{p} \cdot \mathbf{x})}{\hbar} \right]$$

and obtain the relationship between the energy E and momentum $\mathbf{p}$. Consider the corresponding equation for a particle in a non-constant but smoothly varying potential field V :

$$i\hbar \frac{\partial \psi}{\partial t} = \left(-\frac{\hbar^2}{2m} \nabla^2 + V \right) \psi.$$

Starting from the ansatz

$$\psi(\mathbf{x}, t) = a(\mathbf{x}) \exp \left\{ \frac{-i[E t - \varphi(\mathbf{x})]}{\hbar} \right\},$$

where the amplitude a and phase function φ are to be determined, and retaining only the leading-order terms in powers of $1/\hbar$, obtain the eikonal equation

$$H(\mathbf{x}, \nabla \varphi) = E,$$

where H is the classical Hamiltonian. Applying the method of characteristics, show that the particle's phase function can be determined through solution of the classical equations of motion.

8. Consider P-wave rays within an isotropic body. The travel time of a ray between two points $\mathbf{x}_1$ and $\mathbf{x}_2$ can be written

$$T = \int_0^1 \frac{1}{\alpha(\mathbf{x})} \sqrt{\frac{dx_i}{d\gamma} \frac{dx_i}{d\gamma}} d\gamma,$$

where $\gamma \mapsto \mathbf{x}(\gamma)$ is the ray path between $\mathbf{x}_1$ and $\mathbf{x}_2$ parameterised by a generating parameter $\gamma \in [0, 1]$. Regarding this travel time as a functional of the given ray path, Fermat's principle states that the true ray path is a stationary point of this functional. Obtain the corresponding Euler-Lagrange equations, and show that they are equivalent to the Hamiltonian ray tracing equations discussed within lectures. [Hint: it will be useful to express both sets of ordinary differential equations in terms of the arc length along the ray.]

9. Consider an isotropic elastic half-space in which the material parameters vary only with depth $z \geq 0$. Suppose that the P-wave speed α increases monotonically with depth, and consider a ray starting at the surface whose initial tangent vector makes an angle $0 < \theta_0 < \pi/2$ with the z -axis and lies in the (x, z) -plane. Assume that the ray travels monotonically down into the half-space until a depth $z_t > 0$ at which point it is travelling horizontally, and subsequently it turns upwards, returning to the surface in a symmetric manner. By parametrising this ray in terms of the depth co-ordinate z , show that its travel time can be written

$$T = \int_0^{z_t} \frac{2}{\alpha} \sqrt{1 + \frac{dx_i}{dz} \frac{dx_i}{dz}} dz,$$

where $\mathbf{x}$ denotes the horizontal position vector of the ray given as a function of z . Apply Fermat's principle to show that the ray remains in its initial plane, and that along the ray path

$$\frac{\sin \theta}{\alpha} = q,$$

where θ denotes the local angle between the vertical direction and the tangent to the ray, and q is a constant known as the ray parameter. Obtain, in particular, an implicit equation for the turning depth z_t in terms of q . Show that the travel time T and total horizontal distance X of the ray can be obtained through the following integrals

$$T = \int_0^{z_t} \frac{2}{\alpha} \frac{dz}{\sqrt{1 - q^2 \alpha^2}}, \quad X = \int_0^{z_t} \frac{2q\alpha}{\sqrt{1 - q^2 \alpha^2}} dz.$$

10. Consider a plane wave $u_i = f(t - p_j x_j) a_i$ within a homogeneous anisotropic body, where the slowness vector $\mathbf{p}$ and the unit polarisation vector $\mathbf{a}$ satisfy the Christoffel equation $\Gamma(\mathbf{p})\mathbf{a} = \mathbf{a}$, with $\Gamma_{ik}(\mathbf{p}) = \frac{1}{\rho} A_{ijkl} p_j p_l$. The energy conservation law of Question 2 carries over to the linearised theory if T_{ij} and v_i are replaced by their linearised forms, $A_{ijkl} \frac{\partial u_k}{\partial x_j}$ and $\frac{\partial u_i}{\partial t}$, and only terms of second order in the displacement are retained, giving the energy density and energy flux

$$E = \frac{1}{2} \rho \frac{\partial u_i}{\partial t} \frac{\partial u_i}{\partial t} + \frac{1}{2} A_{ijkl} \frac{\partial u_i}{\partial x_j} \frac{\partial u_k}{\partial x_l}, \quad s_j = -A_{ijkl} \frac{\partial u_k}{\partial x_l} \frac{\partial u_i}{\partial t}.$$

- (a) Show that for the plane wave the kinetic and elastic contributions to E are equal, and that $s_j = E v_j$, where the *energy velocity* is

$$v_j = \frac{1}{\rho} A_{ijkl} a_i a_k p_l.$$

Show that $\mathbf{v} \cdot \hat{\mathbf{p}} = c$, the phase speed, and deduce that $\|\mathbf{v}\| \geq c$, with equality if and only if $\mathbf{v}$ is parallel to $\hat{\mathbf{p}}$. Evaluate $\mathbf{v}$ for P- and S-waves in an isotropic body.

- (b) Let $\lambda(\mathbf{p})$ denote the eigenvalue of $\Gamma(\mathbf{p})$ associated with $\mathbf{a}$, so that the corresponding sheet of the slowness surface is $\lambda(\mathbf{p}) = 1$ and the ray Hamiltonian for this sheet is $H(\mathbf{p}) = \frac{1}{2} \lambda(\mathbf{p})$. Show that

$$\frac{\partial \lambda}{\partial p_j} = \mathbf{a} \cdot \frac{\partial \Gamma}{\partial p_j} \mathbf{a},$$

[Hint: differentiate $\lambda = \mathbf{a} \cdot \Gamma \mathbf{a}$, remembering that $\mathbf{a}$ depends on $\mathbf{p}$ but has unit length.] Hence show that $\mathbf{v} = \partial H / \partial \mathbf{p}$, and deduce that the energy velocity is normal to the slowness surface and equal to $d\mathbf{x}/d\sigma$ along a ray.

- (c) An impulsive point source acts at the origin of the body at time $t = 0$. Using ray theory, show that the wavefront at time t is the surface traced out by $t \mathbf{v}(\hat{\mathbf{p}})$ as $\hat{\mathbf{p}}$ ranges over the unit sphere; this is known as the *wave surface*. Describe the geometric relationship between the wave surface and the slowness surface.